

EBENEZER KWAW

MACHINE LEARNING ENGINEER | HEALTH AI AND PRODUCTION ML

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| [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Website](#) |

PROFESSIONAL SUMMARY

Health domain Machine Learning Engineer and public health data specialist with hands-on experience building, evaluating, containerising and deploying machine learning and NLP systems. Skilled in Python, Hugging Face Transformers, FastAPI, Docker, GitHub Actions, PostgreSQL and AWS services, with applied experience fine-tuning transformer models and supporting deployment through SageMaker, Lambda and ECS during a remote ML engineering internship. Brings more than eight years of public health experience in data quality, monitoring, disease preparedness and stakeholder communication, plus deployed projects spanning malaria outreach intelligence, epidemiological surveillance and predictive health analytics. Strong at translating model behaviour and technical trade-offs into practical decisions for multidisciplinary teams.

TECHNICAL SKILLS

Machine Learning and GenAI	Supervised and unsupervised learning, NLP, Transformers, foundation model fine-tuning and evaluation, prompt engineering, classification, DBSCAN, LightGBM, XGBoost
Engineering and MLOps	Python, FastAPI, Flask, Docker, Git, GitHub Actions, CI/CD, REST APIs, model serving, reproducible workflows, technical documentation
Cloud and Data	AWS SageMaker, Lambda, ECS, EC2, PostgreSQL, SQLite, Firebase, SQL, Pandas, NumPy, Scikit-learn, Hugging Face, Gradio, Streamlit
Analytics and Health	Power BI, Plotly, GeoPandas, geospatial analytics, M&E, data-quality validation, DHIMS, Kobo Collect, health information systems

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Thrive Africa Future of Work | Remote Sep 2025 – Dec 2025

- Developed and deployed a four-class customer-support text classifier by fine-tuning DistilBERT and benchmarking it against TF-IDF and Logistic Regression baselines.
- Prepared and analysed unstructured text data, designed evaluation workflows and improved model accuracy and F1 score through iterative fine-tuning and error analysis.
- Worked with foundation model fine-tuning and evaluation workflows and used AWS SageMaker, Lambda and ECS in the internship deployment environment.
- Built a real-time Gradio inference interface with FastAPI and Flask integration options; containerised services with Docker and documented the deployment architecture.

- Contributed through Git branching, pull requests, peer review and CI/CD workflows, and presented model results and trade-offs to technical and nontechnical stakeholders.

District Nutrition Officer and Public Health Officer | Ghana Health Service | Agona East

2021 – Present

- Lead monitoring, analysis and performance review for district nutrition, maternal health, immunisation and disease-preparedness programmes.
- Manage facility-to-district health data workflows, conduct validation and quality reviews, and develop reporting templates and SOPs that improve accuracy and timeliness.
- Coordinate supportive supervision, train health workers and communicate evidence and programme trade-offs to clinical, operational and community stakeholders.
- Contribute to the Disease Emergency Preparedness Committee, district research activities and outbreak-response planning.

District Representative and IPTp Programme Coordinator | RootsLink Africa Malaria

Programme | Ghana

2022 – Present

- Coordinate community malaria interventions, volunteer supervision, behaviour-change communication and monitoring in collaboration with public health stakeholders.
- Built a Kobo Collect monitoring system for real-time outreach tracking, structured reporting and identification of service gaps.
- Produced district reports and implementation evidence for programme planning and engagement with Ghana's National Malaria Elimination Programme.

SELECTED MACHINE LEARNING AND DIGITAL PRODUCT PROJECTS

MalaGuide Malaria Outreach Intelligence Platform | *FastAPI, PostgreSQL, JavaScript, REST API, Render, GitHub Pages*

- Built and deployed an end-to-end health outreach platform that captures community data, updates dashboards and automatically routes High and Medium priority records for follow-up.
- Developed an AI-enabled malaria education assistant using controlled phrase recognition, spelling-tolerant matching, meaning patterns, clarification flows and structured, safety-conscious responses.
- Implemented a FastAPI backend, PostgreSQL persistence, HTTPS REST integration and CORS connectivity with a responsive frontend; tested the data flow from submission to dashboard update.
- Designed the pilot around transparent decision logic and responsible health communication, positioning the tool for future authentication, evaluation and DHIS2 integration.

| [Live](#) |

LocalSignalAI Local Marketing Intelligence Platform | *Product analytics, SEO data, dashboards, API integration, AI-assisted recommendations*

- Developing a decision-support platform for small businesses to consolidate local visibility, competitor activity, review themes, content gaps, lead sources and campaign performance.
- Translating real multi-client SEO, Google Business Profile, paid media and analytics workflows into reusable product features and measurable recommendations.
- Designing the MVP for scalable data ingestion, executive dashboards and AI-assisted interpretation while documenting current free-tier limitations and the production roadmap.

| [Live](#) |

ADDITIONAL REMOTE AND PRODUCT EXPERIENCE

Digital Marketing Specialist | Amplify Marketing Agency LLC | Virginia, USA |

Nov 2025 – Present

Remote

- Manage distributed, cross-functional delivery for multiple US client accounts and translate GA4, Search Console, advertising and local-search data into actionable recommendations.
- Directed campaigns that generated 23 qualified leads in three days and built repeatable reporting workflows across Google Ads, Meta Ads and Google Business Profile data.
- Strengthened product and customer understanding through hands-on work with small-business data, content systems, technical SEO and performance measurement.

EARLIER HEALTH EXPERIENCE

Nutritionist | Samartex Hospital, Ghana Health Service | Samreboi

2018 – 2021

- Delivered clinical and community nutrition services, monitored outcomes and collaborated with multidisciplinary care teams; also supported HIV and ART counselling from 2018 to 2020.

Nutritionist and ART Dispenser and Counsellor | Shama Health Centre, Ghana Health

2016 – 2017

- Provided nutrition counselling, community outreach, ART support and routine health-data collection.

EDUCATION

Bachelor of Science in Community Nutrition | University for Development Studies |

Ghana

2012 – 2016

SELECTED CERTIFICATIONS AND TRAINING

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- Machine Learning and Artificial Intelligence — Thrive Africa and Koforidua Technical University, 2025
 - Data Science and Analytics — Thrive Africa and Koforidua Technical University, 2025

- Advanced Cloud Engineering — Ongoing
- Introduction and Intermediate Python — DataCamp, 2025
- Cybersecurity — Thrive Africa and Koforidua Technical University, 2025
- Front End Software Engineering — Thrive Africa, 2025
- Project Management for Global Health — University of Washington, 2023
- Monitoring and Evaluation for Global Health — University of Washington, 2023

RESEARCH AND PUBLICATION

Assessing the Prevalence of Hypertension and Obesity among Diabetics in Tamale Metropolis — British Journal of Medicine, 2017.

PROFESSIONAL STRENGTHS

Health-domain expertise | Model evaluation | Responsible AI awareness | Technical documentation | Cross-functional collaboration | Stakeholder communication | Research and monitoring | Remote teamwork